PRASANTHKUMAR PONNADA

Machine Learning | Data Science | Python

E-mail: prasanthppk16@gmail.com | M: +91-9000570196

Professional Summary:

AI/ML Engineer with hands-on experience in Python, Machine Learning, NLP, and Generative AI, with a strong academic foundation in Artificial Intelligence. Experienced in developing end-to-end ML solutions including customer sentiment analysis, CSAT prediction, topic modeling, recommendation systems, and AI-powered applications. Proficient in developing, evaluating, and deploying machine learning models with a focus on practical business applications.

Technical Skills:

- **Programming**: Python, NumPy, Pandas, SQL, C
- **Machine Learning**: Scikit-Learn, Classification, Regression, XGBoost, Random Forest, Model Evaluation, Feature Engineering
- **Deep Learning & NLP**: Transformers, BERT, NLP, Sentiment Analysis, Topic Modeling
- **Data Science**: Data cleaning, Statistical Analysis
- **Data Visualization**: Matplotlib, Seaborn, Plotly
- **Cloud**: AWS*, Azure, GCP
- **Deployment & Tools**: Git, GitHub, Gradio, Hugging Face

Projects:

Customer churn rate prediction in Telecom using Machine Learning

Python | Scikit-Learn | Pandas | NumPy | Gradio | Hugging Face

- Developed Machine Learning Model to predict customer churn using customer demographics, service usage and billing information.
- Performed data preprocessing, EDA, Feature engineering and categorical variable encoding to prepare the dataset for modelling.
- Trained and evaluated classification algorithms such as Random Forest, Logistic Regression and XGBoost using Accuracy, Precision, Recall and F1-score.
- Identified key churn driving factors and generated predictions to help telecom companies implement targeted customer retention strategies.

Customer Sentiment Prediction & Personalized Experience Engine for Airlines

Python | NLP | NLTK | Scikit-Learn | BERT | LDA | Gradio | Hugging Face

- **Built end-to-end feedback analytics pipeline**: Processed 1500 passenger feedback records (Text reviews given by passengers) with Python/Pandas and performed pre-processing on feedback.
- **Developed CSAT scoring model:** Engineered features from structured ratings and sentiment scores (using VADER/TextBlob/BERT + Logistic Regression). Predicted passenger satisfaction (High, Low

and Neutral) with ~90% accuracy. Trained and evaluated multiple machine learning classifiers using accuracy, precision, recall, F1-score and support to identify the most suitable predictive approach.

- **Created Recommendation Engine:** Built a recommendation system by using Topic modelling (LDA) to identify recurring service issues such as Food Supply, Flight delay, etc., and maps to actionable improvement strategies.
- **Insights & Business Impact:** Identified the major issue of low CSAT score is Refund issues. Proposed data driven actions may improve CSAT by some extent. Visualized trends with Matplotlib.
- **Deployment:** Integrated the prediction and recommendation pipeline into an interactive Gradio application and prepared the solution for deployment on Hugging Face Spaces.

Fashion MNIST Image Classification using Machine Learning

Python | NumPy | TensorFlow | Scikit-Learn | Keras | CNN | Matplotlib

- Developed an image classification model to automatically identify fashion items form the Fashion MNIST dataset, covering categories such as shirts, shoes, bags and trousers.
- Performed image preprocessing and normalization to prepare grayscale image data for efficient model training.
- Implemented and trained a neural network-based classifier to learn visual patterns and classify fashion images into their respective categories.
- Evaluated model performance using accuracy, loss, confusion matrix and classification metrics to assess classification effectiveness.

Titanic Survival Data Analysis

Python | NumPy | Pandas | Matplotlib | Seaborn | EDA

- Performed Exploratory data analysis on the titanic dataset to understand passenger demographics, travel characteristics and survival patterns.
- Conducted data cleaning and preprocessing including handling missing values, removing inconsistencies and transforming categorical variables.
- Analyzed the relationship between survival and key features such as passenger class, gender, age, fare and family size using statistical analysis and visualizations.
- Derived actionable insights from passenger survival patterns through charts and correlation analysis to identify factors associated with survival outcomes.

Voice Cloning Application using Coqui TTS

Python | Coqui TTS configured models | VITS | XTTS

- Developed a voice cloning application using Coqui TTS to generate natural sounding speech that replicates a target speaker’s voice characteristics.

- Implemented audio preprocessing and voice sample handling to prepare speaker recordings for cloning and speech synthesis.
- Integrated text-to-speech pipeline that converts user-provided text into synthesized speech using the cloned voice.

Static Website Deployment through AWS

AWS | Azure | GCP | S3 | EC2 | VM

- Developed and deployed a static website on AWS using HTML with a focus on accessibility and responsive design.
- Configured Amazon S3 for static website hosting including bucket setup, file management, and public access policies.
- Configured domain and HTTPS access to provide a secure, and scalable web hosting solution on AWS, Azure, GCP.

Professional Experience:

CAE Senior Anlayst

Satyam Venture Engineering Services | Hyderabad

Feb 2022 - Present

- Performed Finite Element Modelling and Analysis activities using engineering simulation tools.
- Developed and reviewed simulation models while ensuring accuracy and adherence to project requirements.
- Collaborated with cross-functional teams to resolve technical issues and deliver projects withing defined timelines.
- Received Star of the Month recognition for consistent commitment to timely and error-free project deliveries.

CAE Engineer

Xitadel CAE Technologies India Pvt. Ltd. | Bangalore

Nov 2018 – Jan 2022

Education:

- **Associate of Science** in Artificial Intelligence

  Saras AI Institute**, Salt Lake City, UT, USA.* | ***CGPA: 3.5 / 4

  Relevant coursework:

  Machine Learning | Deep Learning | NLP | DSA | Gen AI | Statistics | Maths for AI

- **Master of Technology** in Thermal Engineering

  Gayatri Vidya Parishad College of Engineering**, Visakhapatnam, A.P.* | ***Percentage: 69.17%

- **Bachelor of Technology** in Thermal Engineering
  Gudlavalleru Engineering College*, Krishna, A.P.* | ***Percentage: 72.15%***

Certifications:

- Machine Learning – [Saras AI Institute]
- Python for Data Science – [Saras AI Institute]
- AWS Cloud Practitioner

Achievements:

- ACE Performer – Recognized for achieving highest score through dedication perseverance and academic excellence in few courses such as Powers of AI, AI in Cloud and Communication styles.
- Successfully developed and deployed multiple AI/ML projects as part of academic and practical coursework.
- Participated in AI/ML technical projects involving NLP, Machine Learning, Recommendation systems and Application deployment.
- Start of the Month – Recognized for commitment to timely and error-free delivery of modelling projects.

Place: Hyderabad
Date: 05-October-2026 [Prasanthkumar Ponnada]